

1. During the gradient calculations

Given:

$$G_x = 24$$

$$G_y = 18$$

$$\begin{aligned} \text{Gradient magnitude } |G| &= \sqrt{G_x^2 + G_y^2} \\ &= \sqrt{(24)^2 + (18)^2} \\ &= \sqrt{576 + 324} \\ &= \sqrt{900} \end{aligned}$$

$$|G| = 30$$

$$\text{Gradient direction } (\theta) = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{18}{24} \right)$$

$$\theta = \tan^{-1} (0.75)$$

$$\theta = 36.86^\circ$$

2.

A row of gradient

Pixel	P1	P2	P3	P4	P5
Magnitude	25	50	90	60	30

For a pixel to survive NMS, its magnitude must be greater than or equal to its two neighboring values.

check P1:

$$P1 = 25$$

$$\text{neighbour } P2 = 50$$

$$25 < 50 \quad P1 \text{ is suppressed}$$

check P2:

$$P2 = 50$$

Neighbours

$$P1 \text{ \& } P3$$

$$\begin{array}{cc} \downarrow & \downarrow \\ 25 & 90 \end{array}$$

$$50 > 25 ; 50 < 90$$

$\therefore P2$ is suppressed.

check P3

$$P3 = 90$$

Neighbours

$$P2 = 50; P4 = 60$$

$$90 > 50; 90 > 60$$

P3 is a local maximum

check P4

$$P4 = 60$$

Neighbours

$$P3 = 90$$

$$P5 = 30$$

$$60 < 90; 60 > 30$$

P4 is suppressed.

check P5

$$P5 = 30$$

$$\text{Neighbour } P4 = 60$$

$$30 < 60$$

P5 is suppressed.

Pixel	P1	P2	P3	P4	P5
Original	25	50	90	60	30
After NMS	0	0	90	0	0

$\therefore$ only P3 with magnitude 90 is retained as an edge pixel.

3.

Given:

$$\text{High Threshold} = 100$$

$$\text{Low Threshold} = 50$$

The gradient magnitudes are

Pixel	P1	P2	P3	P4	P5	P6
Magnitude	135	90	45	120	70	20

$$1) P1 = 135$$

$$135 \geq 100$$

strong edge

$$2) P2 = 90$$

$$90 < 100$$

weak edge

$$3) P3 = 45$$

$$45 < 50$$

No-edge

$$4) P5 = 70$$

$$70 > 50$$

weak edge

$$5) P6 = 20$$

$$20 < 50$$

No-edge.

strong edges:
P1, P4

weak edges:
P2, P5

No-edge:
P3, P6

4.

Given	P1	P2	P3	P4	P5	P6	P7
Pinel							
Magnitude	130	95	40	115	65	25	105

P1
 $130 \geq 100$
strong

P2
 $95 < 100$
weak

P3
 $40 < 50$
No-edge

P4
 $115 \geq 100$
strong

P5
 $65 > 50$
weak

P6
 $25 < 50$
No-edge

P7
 $105 \geq 100$
strong

strong:
P1, P4, P7

weak:
P2, P5

No-edge:
P3, P6

After Hysteresis:

P2 is connected to strong edge $P1 \rightarrow P2$ is retained
P5 is not connected to any strong edge $\rightarrow$ P5 is removed.

$\therefore$ P1, P2, P4, P7.

Pinel	Final
P1	Edge
P2	Edge
P3	Non-Edge
P4	Edge
P5	Non-Edge
P6	Non-Edge
P7	Edge